



Practice Assignment 8

Practice Assignment – Quarter 2, 2023-2024

Course Name: Database

Course Code: CSE 301

Student's Full Name:

Student ID:

Overview:

Part 1: Essay question

1. Find Entities, Relationships and Draw ER-Diagram.
2. Find the canonical cover. (Do exercises again)
3. Write algebra relationships.

Part 2: Practice questions

4. Create constraints for database.
5. Write SQL queries.

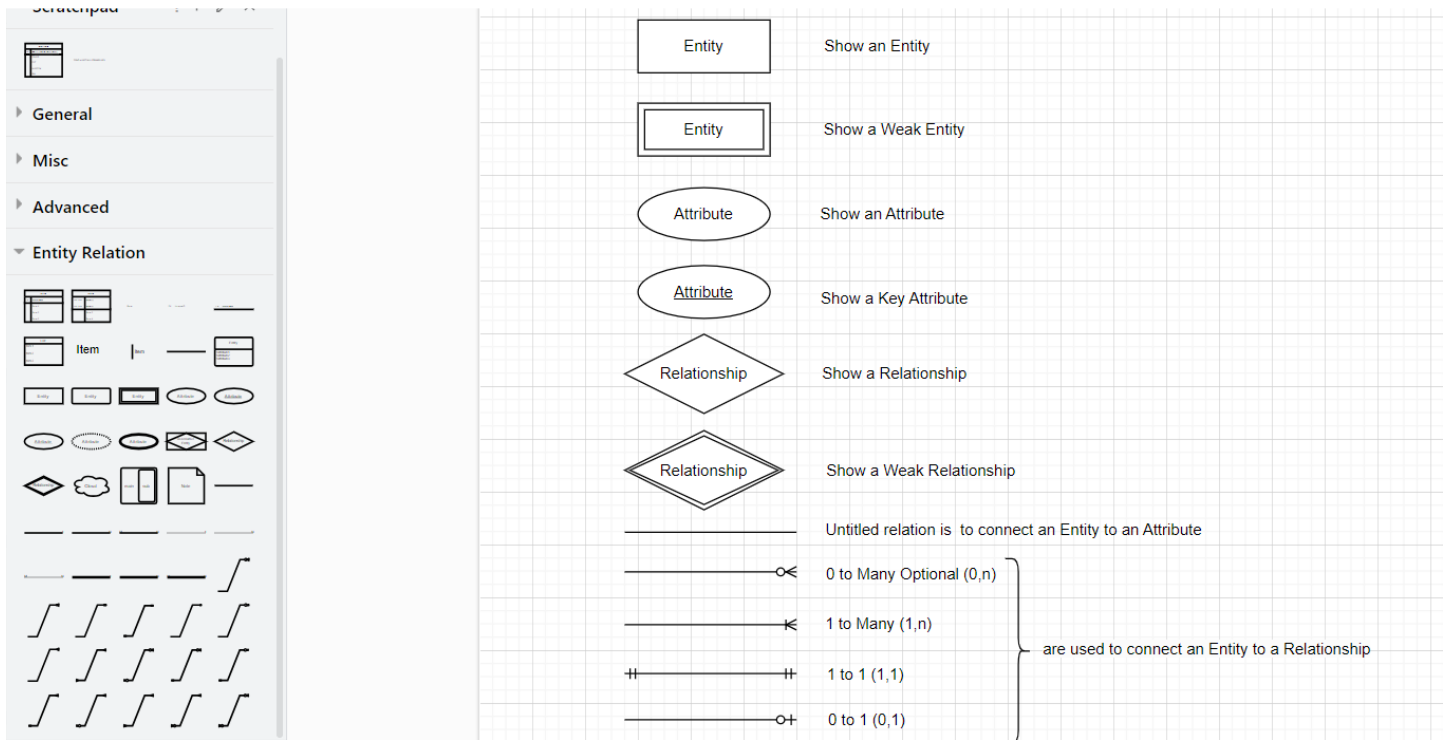
Problem 1: Building an ER model and relational data model for 1 hotel needs to store information as follows:

The hotel has many rooms, each room has room code (R101, R102, ...), room name (sunflower room, jade room, ...) Each room is located on 1 floor, floor information has floor code (F01, F02, ...), floor name (1st floor, 2nd floor, ...) Each room has a price corresponding to whether guests stay by week or by day. Each guest when arriving at the accommodation will have an identity card number, full name and corresponding phone number. There is also service information attached for each room: service code, service name and corresponding amount for each service when used 1 time. For example, service code: S01, service name is "Laundry", the amount is 50000.

Room payment invoice information for customers will have information: Invoice code, identity card code, full name of the customer, number of days of stay (arrival and departure dates), in which room, total amount to pay for the room. If guests stay for 1 week and 2 days, the week will be calculated according to the weekly unit price plus 2 odd days will be calculated according to the daily unit price.

The invoice also stores information: service name and total service amount. The total amount of the guest to be paid (the sum of the total room amounts and service fee).

Optional symbol	○
One symbol	
Many symbol	≧



Problem 2: Writing algebra relationships using Database “managementproject” in Lab 2 according to questions below:

- Find employees who work in room 4.
- Find employees with salaries above 30000.
- For each department, indicate the department name and room location.
- The average salary of all female employees.
- Find the names and addresses of all employees of the "Nghien Cuu" department.